An Innovative Approach to Involve Students with Learning Disabilities in Intelligent Learning Systems

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Abstract. Innovations in healthcare education have fostered the development of assistive technologies which have contributed to making the learning environment and information more accessible to learners with learning difficulties. Especially, Intelligent Tutoring Systems (ITS) have proven the benefits of personalized learning. One of the distinctive features of the intelligent system is its Learner Model (LM). In this paper, we propose a LM that takes into consideration individual differences and tends to adapt system parameters for learners with Specific Learning Disabilities (SLDs) before the session begins. The purpose of this research is the development of conceptual bases and a constructional approach of a cognitive LM founded on differentiation. However, developing ITS requires many resources and information about learners as well as it demands a long time to build. To facilitate the design process, we propose a LM, called UPCLEE that takes into account several dimensions of learner's profile. This research demonstrates how the proposed model's levels are taken into account in the design of ITS and how it can be used to provide personalization in computer-based educational systems. In this study, we collected observations, interviews, and surveys and we used a grounded theory approach to develop our LM, then we conducted a user study with 12 students (4 with SLDs) to validate our proposed model.

Keywords: Healthcare education · Information and Communication Technology (ICT) · Specific Learning Disabilities (SLDs) · Intelligent Tutoring System (ITS) · User experience · Learner Model (LM)

1 Introduction

Students with Specific Learning Disabilities (SLDs) often have many kinds of difficulties that lead to academic failure or low achievement. The most common types of learning difficulties involve problems with reading, writing, math, attention, listening, and speaking, such as dyslexia, dyscalculia or dysphasia. SLDs refers to a disorder in one or more of the psychological processes involved in understanding or using spoken

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or written language. The term SLDs does not involve learning problems that are primary the result of visual, hearing disabilities or mental retardation.

Dyslexia is one of the most common learning disabilities, it is a disorder manifested by difficulties in reading. According to [1], the percentage of dyslexics among the normal population is around 7%.

Recently, studies are focusing on the potential benefits of Information and Communication Technology (ICT) use in healthcare education to improve and develop interactive experiences that can assist and help students with SLDs [2]. For instance the work of [3] about the use of assistive mobile applications for individuals with dyslexia aimed at reeducating and monitoring the learning process, or the work of [4] about the inclusion of individuals with learning difficulties in virtual learning environment.

Although there are a variety of research within the areas of SLDs, ICT use, and a wide range of modeling approaches, there still seems to be a research gap. However, the effectiveness of several applications has not been demonstrated. Plus, there is no unified strategies in designing e-learning systems for individuals with SLDs neither guidance for choosing a model suitable for particular learner. Research is, therefore, needed on the instructional design side of ICT for individuals with SLDs.

LM is a key element in adaptive learning systems. However, the challenge lies in the fact that individuals with SLDs have specific cognitive ability characteristic and different learning needs. Although extensive research has been carried out on learner modeling and intelligent e-learning systems, there have been no studies found that investigate LM for individuals with SLDs. Hence, this present study took the initiative to explore the learning experiences of students with SLDs so as to propose an innovative approach based on differentiation, with the aim to design a LM that takes into account a several dimensions of learner's profile. Therefore, we decided to investigate the possibilities offered by advanced technologies, to develop a conceptual base and a constructional approach of a cognitive user model, which we called UPCLEE.

The rest of this paper is organized as follows. The next section covers related work, followed by our proposed approach in Sects. 3 and 4 presents the user study and conclusion and future work in Sect. 5.

2 Related Work

2.1 Communication Technology for Learners with SLDs

Some researchers have been interested in assistive e-learning systems, for instance the work of [5] that proposed an assistive learning environment to enhance the learning experience of learners with SLDs in their academic life. The aim of this assistive system is to transform the content in an acceptable way by learners with SLDs. The architecture of the framework contains 6 major modules as follows: Content repository, learner profile, assistive learning engine, transformation base, learners' goal and monitoring module. However, the main disadvantages of the system were that it was focused only on one major barrier of dyslexic learners, which is reading activity. Therefore, the system seems to be not useful for all types of dyslexic learners, because it does not differentiate between types of dyslexia.

Related studies [6] about the inclusion of students with learning difficulties in ICT use, proposed an adaptive mobile learning system based on the analysis of dyslexic students' learning style preferences. The outcomes stated that learners with dyslexia have strong preferences for a specific learning style, compared to learners from control group that have mild learning style preferences. Although, research solutions developed by these researchers were significant and adequate, most of them focused only on fundamental literacy skills, weaknesses and difficulties. However, they seem not helpful for all individuals with SLDs. Especially because the degree of their learning disabilities differs. Thus, there is still a lack of integrated tools that provide individuals with learning difficulties with the appropriate set of adjustments according to a diversity of individual learning needs.

2.2 Learner Model (LM)

LM has triggered interest from many researchers. Genetic Graph method [7] is the first approach to a learner-based modeling, the system analyzed the requisite skills for playing a game into atomic components and created links between these components. Recent research work [8] has been done on implicit user modeling based on artificial intelligence. Accordingly, implicit data captured during the interaction with web systems and statistical methods and clustering techniques were performed. The outcomes reveal a strong significant relationship between the users' cognitive styles and preference toward a specific type of learning content.

Many studies have proposed solutions to overcome the literacy barriers of students with learning disabilities. Some researchers have been interested in e-learning systems. One example is [9] who developed a software system for reading and writing called iLearnRW for students with dyslexia. Another example is [10] who proposed a LM technique for individuals with learning difficulties in virtual environments, or the work of [4] who studied the importance of personalized assistive tools for learners with SLDs. However, LM plays a principal role in learning and it has important implications for e-learning systems. LM can be defined as the process of creating a model of the learner to personalize a system. Generally, LM represents essential information about each learner and control her/his goals, the content she/he views, and the manner in which she/he navigates in the virtual environment. According to [11], LM is a representation of static and dynamic information about an individual that is utilized by the adaptive interactive system to provide adaptation effects. LM can be based on explicit or implicit information. Explicit collection of learner's information is provided directly by the learner through Web registration forms or questionnaires, while implicit collection of learner's information is extracted automatically by systems to presume learner's characteristics and it is obtained by tracking the learner's navigation behavior by interacting with the system.

3 Our Proposed Model UPCLEE

Learners are characterized by different learning styles, abilities, and interests. Differentiated instruction is a term that has been defined in the field of education as a teaching theory that implements varied instructional approaches and methods in order to be adapted according to learners of different ages, abilities and learning skills. So as to attain their goals in different ways and also to make learning interesting and engaging [12].

In intelligent learning systems, LM can contain different information depending on what we are trying to model, especially what characterizes our target learners. It represents what the system knows about the learner, this information could be cognitive, behavioral or psychological. LM can relate to one or more aspects of the learner such as: misconceptions, problem-solving speed, motivation, metacognitive aspects and working memory capacity. However, to design LM we should know the characteristics of each learner. Hence, the challenge lies in the fact that individuals with SLDs have specific cognitive ability characteristic and different learning needs. Therefore, our proposed model is founded on differentiation and it tends to analyze and adapt the learning environment that takes into consideration learners characteristics. This model takes into account different dimensions of learner profile. The main objective is to provide a structured learner profile and provide learning systems with relevant information to adapt the learning to the specificities of learners with SLDs. The idea behind this model is to introduce a system that pays attention to the specific needs of learners with SLDs, assesses, diagnoses their problems and provide help.

3.1 Levels of Our Proposed Model UPCLEE

The proposed model addresses different levels that serve to better understand the difficulties of learners with SLDs. UPCLEE (User profile, Preferences Profile, Cognitive Profile, Learning Profile, Errors Profile, and Experience Profile) focuses particularly on 6 levels wish are based on the researched theories in the respective domain as described as follows:

- Level 1 User Profile

Demographic Data: This is a static level that represents demographic information that represents the learner in terms of personal information which includes a set of attributes of identification, such as name, surname, age, gender, etc.

- Level 2 Preference Profile

Learning preference or learning style refers to the preferential way in which the student absorbs, comprehends and retains information. There are numerous learning models and theories that attempt to clarify and explain the learning styles differences, such as Kolb's learning styles model [13], Felder and Silverman [14], and VAK learning style model [15].

Previous work [6], investigated learning styles preferences of students with specific learning disabilities. Researchers administered VAK and Honey & Mumford questionnaires to 28 Moroccan students with ages between 8 and 10 years old, 8 students

with dyslexia and 20 students without SLDs to compare student's preferred learning style. The outcomes suggested that individuals with SLDs have strong learning style preference. In the same study they proposed an adaptive m-learning game, which matches directly with students learning styles through the use of advanced digital technology. The study confirmed that when students have an understanding of their learning style dominance, they could be empowered to learn in terms of their strengths. Other study was carried out by [16] to investigate whether teaching to students with dyslexia's preferred learning styles would improve their performance and attainment in both literacy and numeracy, the researcher conducted a user study with seven British students aged between 7 and 8 years, using both qualitative and quantitative methods. The outcomes showed an improvement in performance and attainment in spelling and numeracy. It also indicated that teaching and learning styles influence the quality of learning, assuming that when students with dyslexia are taught to their strengths and given more ownership over their learning, then their whole school experience may be improved.

On the basis of the evidence currently available, it seems fair to suggest that providing an adaptive learning environment that takes into account learners' learning style could positively affect their performance. Further, if it is confirmed that the students are taught using their preferred learning styles, they might improve their attitudes to learning and behavior. Therefore, the objective of this level is to meet the needs of learners and determine their preferences for learning materials.

- Level 3 Cognitive Profile

Prior identification of the learner's needs and cognitive characteristics is an important aspect of the design process of intelligent e-learning systems. Cognitive styles are information processing habits representing the learner's typical style of perceiving, thinking, problem solving and remembering. In fact, each individual has his/her own styles for collecting and organizing information into knowledge [16]. However, to provide systems with this kind of information, at least a user profile is needed to gather user data. In a study conducted by [4], who investigated the use of dyslexic's cognitive styles and learning preferences as determinant factors for personalization. The authors used the interactions between cognitive traits model (CTM) [17], Felder-Silverman learning style (FSLSM) [13] dimensions, the dual coding theory [18] and VAK learning style model [15] to reveal some possible links between dyslexic's cognitive styles and learning preferences. The outcomes showed that the use of the cognitive style mapping for students with SLDs represent an important element in LM.

In other study carried out by [19] to investigate the benefits of incorporating learning styles and cognitive traits in virtual educational systems. The researchers conducted an experiment with 297 students was performed to improve student modeling, relationships between learning styles and Working Memory Capacity (WMC). The results showed the existence of a relationship between learning styles and cognitive traits.

Therefore, in this current investigation we add the cognitive profile level that represents cognitive style of learners with SLDs. This level emphasizes the importance

of individual cognitive characteristics such as WMC, inductive reasoning skill, and associative learning skill.

- Level 4 Learning Profile

This level focuses on defining learning objects (LO) according to the learner's state of knowledge in order to provide optimal conditions for learning. LO is a small reusable group of contents to provide tutorials to the learner. There are numerous specifications and standards to develop LO such as IEEE LOM, IMS Learning Resource, and IMS CP.

In a study conducted by [19] they developed a personalized e-learning system for students with SLDs. It is an assessment model for alphabet learning with learning objects that includes: (i) Assessment model (ii) Ontologies and architecture and (iii) Knowledge-base and learning object. The researchers claimed that this system is useful for students who face dyslexia in elementary school mainly in the age group of 6 and 9 years old. The outcomes indicate the effectiveness of the use of e-learning methods, such as cognitive trait, ontologies and LO.

- Level 5 Errors Profile

Our approach assumes that each learner is unique, in that, the virtual learning environment must encourage and support individual differences [20]. This level identifies the particular errors of each learner. The phonological awareness that can encircle errors during reading and writing, errors in problem solving and repetitive errors. Indeed, the identification of the errors specific to learners with SLDs, makes it possible to identify the fields in which the learner has low performance and to propose to her/him aids in the deficit areas. The importance of this level was confirmed by many research studies across different languages and cultures. One example is the work of [21], about spelling errors of adults with dyslexia, the researchers conducted a user study with 100 students with dyslexia and 100 matched control students, all participant had Dutch as mother tongue. The results indicate 3 main error categories: phonological, orthographic, and grammatical errors. They indicate, also that higher-education students with dyslexia made on average twice as many spelling errors as the controls.

Other examples can be found in the study done by [22], who investigated reading and reading-related skills of 15 French-speaking adults with dyslexia compared with chronological-age controls and reading-level controls. The results reveal that adults with dyslexia exhibited lower phonological reading-related skills than control group only and were better than reading-level controls on the rapid automatic naming, also, a deficit in the sub-lexical reading procedure was observed when compared to reading-level controls. Also, the work of [23] about Chinese speaking aphasic patients reveals subtypes of acquired dyslexia and dysgraphia in Chinese, the outcomes assume that the language environment, and specifically the type of script used to read and write, play an important role in determining the phenotype of dyslexia in Chinese. Another similar study is the work of [24], who investigated the experience of Arabic learners (group of 32 students) when reading online text. The outcomes indicate that dyslexic learners made significantly more spelling errors in reading Arabic online text compared to non-dyslexics. This comparison between learners with dyslexia and chronological-age controls has yielded insights into the need to consider features of the Arabic language

and the need to provide the appropriate set of adjustments according to the diversity of user differences.

The evidence presented in this level suggests that the identification of learner's errors in reading and writing is very important in learning environment and it should not be ignored.

- Level 6 Experience Profile

This level is interested in adapting the presentation of the knowledge and helping the learner navigates through the graph composed by all the components, taking into account the progress of the learner according to her/his experiences. Learner experience can be defined as an internal state (emotions, motivation, and mood) and sensations about systems' characteristics (complexity, usability, functionality) that learners may experience during their interactions with the learning system. Confirming to [25] the satisfaction of psychological needs such as competence, relatedness, autonomy, meaning, security are the principal sources of positive experience with interactive technologies.

Studies that investigated the experience profile can be found in the work of [26], who designed a model for serious game from a user experience perspective for deaf children experiencing difficulty in literacy learning. The proposed model includes four elements: user profile, learning strategies, serious games and requirements. The researchers conducted a user study with 6 deaf children from Mexico, aged between 12 to 15 and have problems in learning literacy. They used research techniques, including activities, such as observations, interviews with the teachers, and questionnaires so as to discover the needs and level of literacy learning. The researchers claimed that the proposed model might be useful for identifying the relevant needs to support deaf children and their teachers and they assumed that these four elements are important aspects to be considered in designing serious games for children with learning difficulties. Figure 1 illustrates the levels of our proposed model UPCLE.

3.2 UPCLEE Life-Cycle

The initialization of user profile can be considered as a starting point of the life-cycle of users with special needs. This step collects information on users and built a model of the learner based on UPCLEE's levels. Generally, there are two ways to obtain data, so that the system can initialize the learner model: implicit or explicit questions such as tests and questionnaires. Then, the construction of the learner model based on the gathered data. However, there exist numerous models to construct a learner model, for instance, stereotypes, overlay and buggy model.

Stereotypes make it possible to group all learners of an adaptive system into several groups according to certain characteristics that are generally shared. Stereotypes are a set of frequent characteristics of the user they represent a category or group of learners. There are two types of stereotypes; fix and default. In the fix type, the learner is assigned to a predefined stereotype at the abstract level, while the default type is more flexible. First, the learner is assigned to the initial stereotype. This means that the initial stereotype has a default value. Then, the system will observe the learners and gather

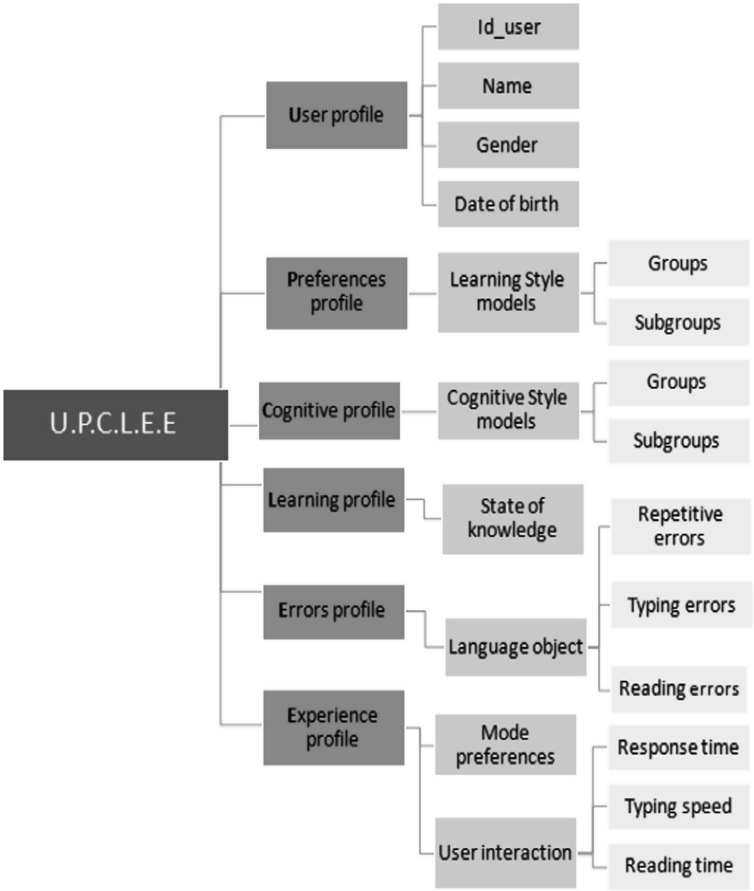


Fig. 1. UPCLEE levels.

their data in learning process. Finally, the system changes the initial stereotype to a new, more precise stereotype.

The overlay model is based on the fact that the learner model is a subset of the domain model. Buggy model is based on a list of potential errors that learners could commit by adding to the knowledge of the expert a library of misconceptions (bug library).

The construction is followed by the exploitation and updates. Finally, the reutilization of profiles that consists in the use of learner profiles to customize the learning activities to come. Figure 2 illustrates the life-cycle of our proposed model UPCLEE and the system prototype.

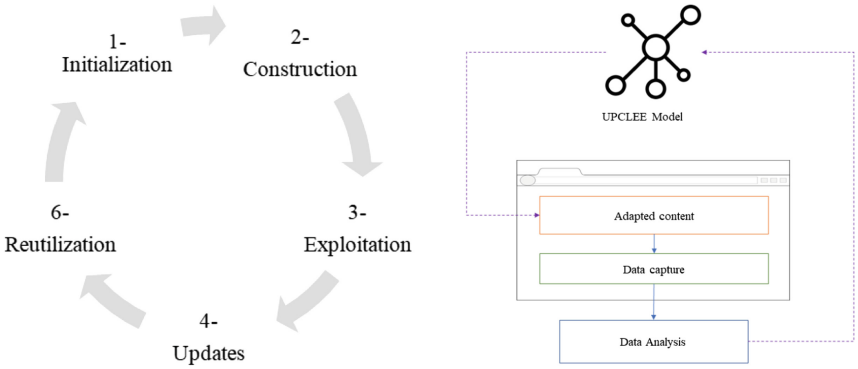


Fig. 2. The proposed adaptive e-learning system (a) UPCLEE life-cycle (b) System prototype.

4 User Study

In order to validate our proposed model, we conducted a user study with 12 students (4 with SLDs) to evaluate the importance of UPCLEE level's. In this present experiment, we focused on two UPCLEE levels, Errors profile and Experience profile. We compared the profile of students with and without SLDs.

Errors profile: the main objective of this level is to collect and identify student with SLDs errors in writing and reading in Arabic, then compare it with the control group. All participants were asked to read an online text and type it, the text contains about 80 words. The score was calculated as follows; how many misspellings per word. It is also possible that one single word contains more than one error; it can have different types of spelling errors.

Experience profile: the main objective of this level is to study the experience of students with SLDs in e-learning environments. For that, we focused on time spending in virtual environments, we used an educational game developed by [5] and we calculated the time spending in the game (seconds).

4.1 Participants

The sample of this study is composed of 12 participants (n = 12) and divided into two groups, the experimental and control group. The experimental group consists of 4 participants aged between 8 and 12 years. They were selected by the head of the phonology department of language-health center - El Jadida Morocco. The control group is composed of 8 children with the same age, they were selected from primary schools from El Jadida city. According to their medical records, they had no prior diagnosis of learning disabilities. All participants were native Arabic speakers. Table 1 presents the distribution of participants in terms of gender and group.

Table 1. Distribution of participants in terms of gender and group

	Experimental group	Control group (NoSLDs)[a]	Total
Girls (n)	01 (with Dyslexia and Dysorthographia)	04	05
Boys (n)	03 (one with Dyslexia and Dysorthographia and two with Dyslexia)	04	07
Total (n)	04	08	12

[a] No Specific Learning Disabilities

4.2 Results

The statistical analysis was performed using R software, version (3.5.0), with condition of $p < 0.05$ for significant results. Table 2 presents results of t-test.

Table 2. Results of t-test and descriptive statistics for errors profile and experience profile by groups.

	Experimental group		Control group (NoSLDs)		95% CI for mean difference		
	M	SD	M	SD	t	df	p
Errors profile							
Reading errors	38.75	09.32	04.75	02.43	7.172	3.20	0.0044**
Writing errors	66.75	11.87	03.62	03.02	10.46	3.19	0.0014**
Experience profile							
Time spending in the game	1005.0	57.44	615.0	146.0	6.59	9.809	<0.001***

* *P < 0.05*
** *P < 0.01*
*** *P < 0.001*

Primary results reveal that students with and without SLDs have different profile which confirms the importance of UPCLEE levels in e-learning environments.

5 Conclusion and Future Work

The growing importance of advanced technologies applied to healthcare education offers to students with various backgrounds the opportunities to develop their learning skills and overcome their difficulties. This work, showed how new initiatives in the field of education at the crossroads of healthcare and technology can be shaped and implemented in practice.

In this study, we proposed a cognitive LM for learners with SLDs, which we called UPCLEE. The model takes into consideration the characteristics of each learner. It collects 6 levels of learner profile: User profile, Preferences Profile, Cognitive Profile, Learning Profile, Errors Profile and Experience Profile. The primary evaluation confirms the importance of UPCLEE levels in e-learning systems.

However, at this stage this kind of initiative can only be achieved through multidisciplinary collaborations with many functional, technical and financial means. Therefore, the exploitation of such a model seems to be the next step, to realize a first functional prototype of our model. Clinical studies can then be carried out to prove the effectiveness of the proposed approach so as to support students with learning disabilities and other neuropsychological disorders.

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